SOFTWARE ENGINEERING AND DESIGN PRINCIPLES
(CS311L)

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AI CHATBOT FOR COLLEGE HELPDESK

UML Sequence and Activity Diagram Modeling

1. Aim

To model the major scenarios of the AI Chatbot for College Helpdesk using UML Sequence Diagrams to represent message flow between actors and system components, and an Activity Diagram to summarize the overall query-handling workflow.

2. Major Scenarios

The following scenarios are selected with reference to the supplied use-case diagram and the Software Requirements Specification (SRS).

Scenario	Participants	Purpose
1. General College Query	User, Chat Interface, Chatbot/NLP, Knowledge Base	Processes a general query and displays a matching FAQ or college-information answer.
2. Personalized Student Information	Student, Chat Interface, Authentication Service, Chatbot, ERP/SIS	Authenticates a student before retrieving personal academic or fee information.
3. Unresolved Query / Ticket Escalation	User, Chatbot, Ticket Service, Helpdesk Operator, Notification Service	Creates and routes a support ticket when the chatbot cannot resolve a query.

3. Sequence Diagram 1: General College Query

This scenario represents a student, faculty member or visitor asking a general question. Authentication is not required for general college information.

1. The user enters a query through the chat interface.
2. The interface sends the query text to the chatbot/NLP component.
3. The chatbot identifies the intent and extracts relevant entities.
4. The chatbot searches the knowledge base for a matching FAQ or answer.
5. The knowledge base returns the matching answer to the chatbot.
6. The chatbot formats the response and the interface displays it to the user.

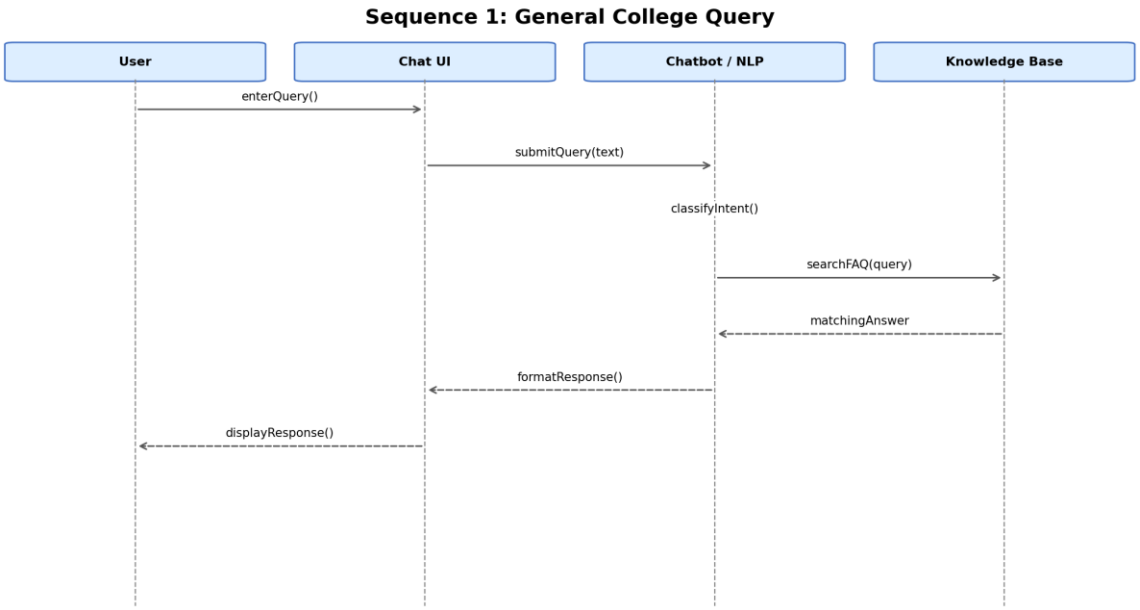


Figure 1. Sequence diagram for a general college query.

4. Sequence Diagram 2: Personalized Student Information

This scenario shows how a student accesses personal academic or fee information. The system must verify the student's identity through login/OTP before retrieving data from the college ERP/SIS.

7. The student requests personalized information such as attendance or fee status.
8. The chat interface initiates login/OTP verification.
9. The student submits credentials or OTP to the authentication service.
10. After successful verification, the chatbot requests the required data from ERP/SIS.
11. ERP/SIS returns the requested student data to the chatbot.
12. The chatbot formats the result and the chat interface displays it to the student.

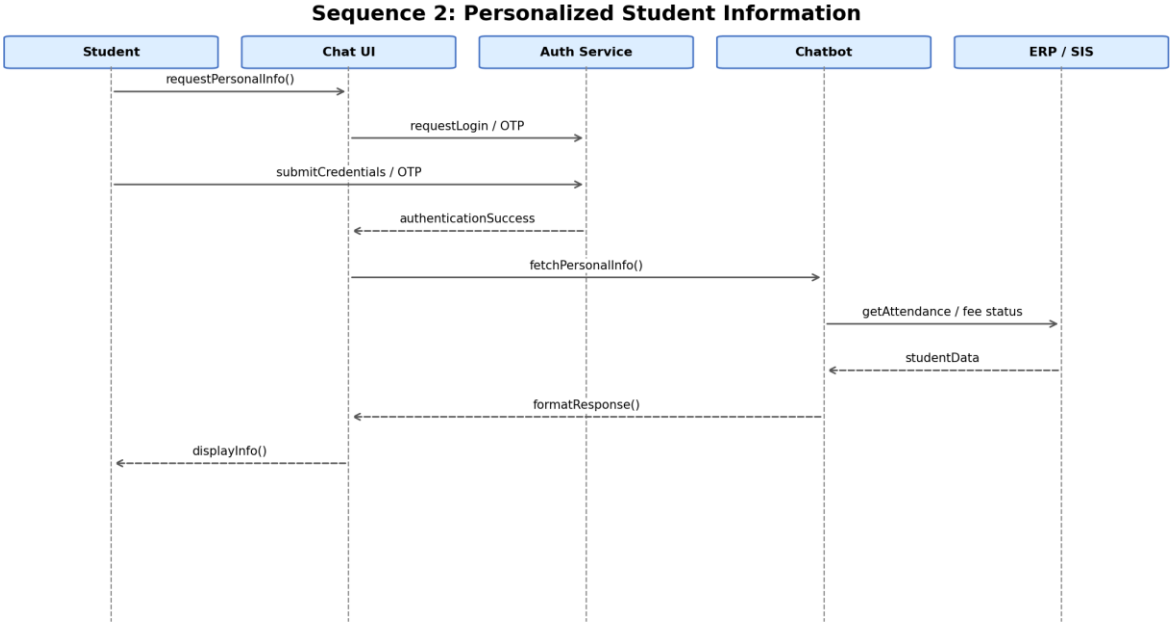


Figure 2. Sequence diagram for personalized student information.

Alternative flow: If authentication fails, access to personal information is denied and the student is asked to retry. If ERP/SIS is unavailable, the system displays an error and may offer helpdesk escalation.

5. Sequence Diagram 3: Unresolved Query and Support Ticket

This scenario represents the escalation of an unresolved query to a human helpdesk operator. The ticket includes the query summary and relevant conversation context.

13. The user submits a query that the chatbot cannot resolve.
14. The chatbot sends the query summary and conversation context to the ticket service.
15. The ticket service creates a unique ticket ID and assigns a category.
16. The ticket is routed to the relevant helpdesk operator.
17. The notification service sends the user a ticket confirmation.
18. The operator reviews the context and works on the query.
19. The ticket service updates the ticket status and the user can be notified.

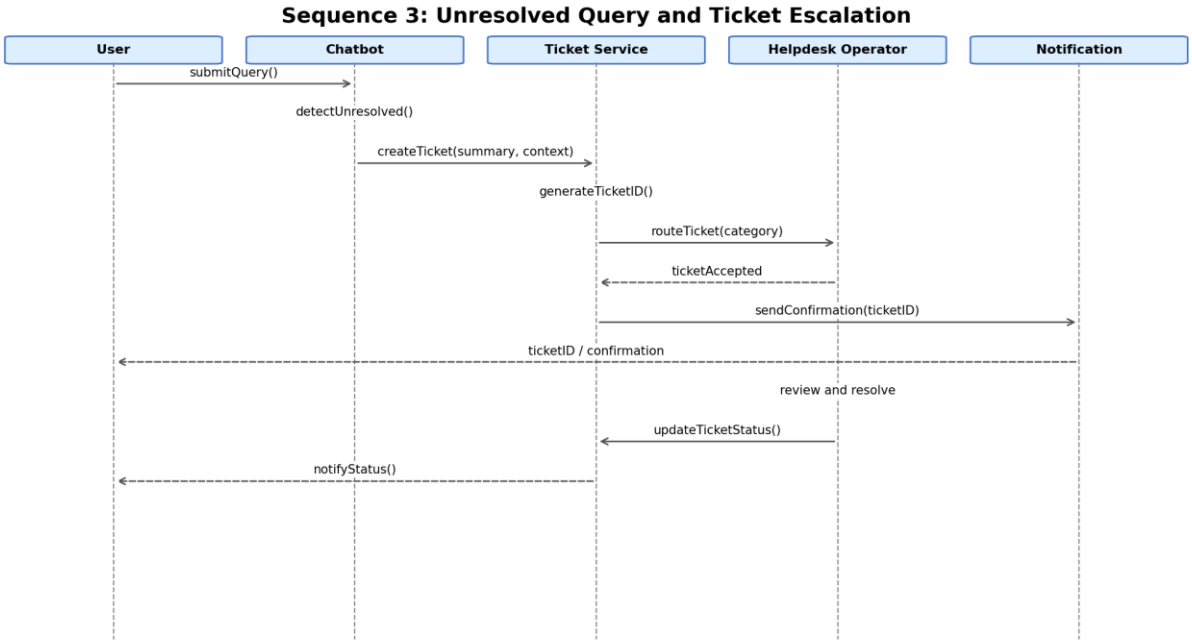


Figure 3. Sequence diagram for ticket creation, routing and status update.

6. Activity Diagram: Query Handling Workflow

The activity diagram summarizes the overall workflow followed by the chatbot. It includes intent identification, the decision between personal and general information, authentication, ERP/SIS retrieval, response display, and escalation when the query remains unresolved.

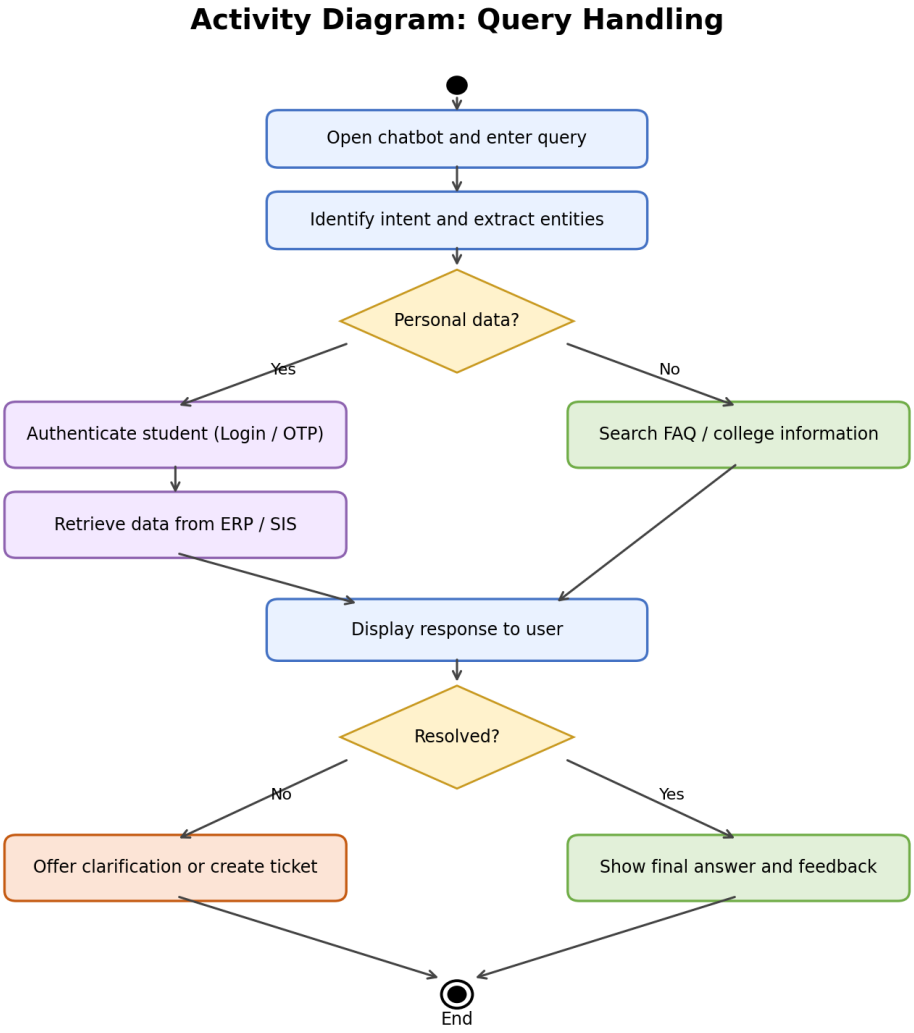


Figure 4. Activity diagram for query handling.

7. Design Notes

- General college information can be accessed without login, while personal academic and fee information requires successful student authentication.
- ERP/SIS is an external college system used to retrieve student-specific data.
- The knowledge base stores FAQ entries and general college information used by the chatbot.
- Unresolved queries can be escalated through support tickets routed to the appropriate helpdesk department.
- Helpdesk Operators can monitor conversations and update ticket status. Authorized Admin users can manage the knowledge base and view analytics.

8. Conclusion

The sequence diagrams model the message flow for a general college query, authenticated retrieval of personalized student information, and escalation of an unresolved query through a support ticket. The activity diagram summarizes the overall query-handling workflow. These models clarify the order of interactions and the responsibilities of the user, chatbot components, college systems and helpdesk staff.

9. References

1. Software Requirements Specification for AI Chatbot for College Helpdesk, Version 1.0.

2. Supplied Use Case Diagram and Use Case Specifications for AI Chatbot for College Helpdesk.

3. UML Sequence Diagram and Activity Diagram notation.